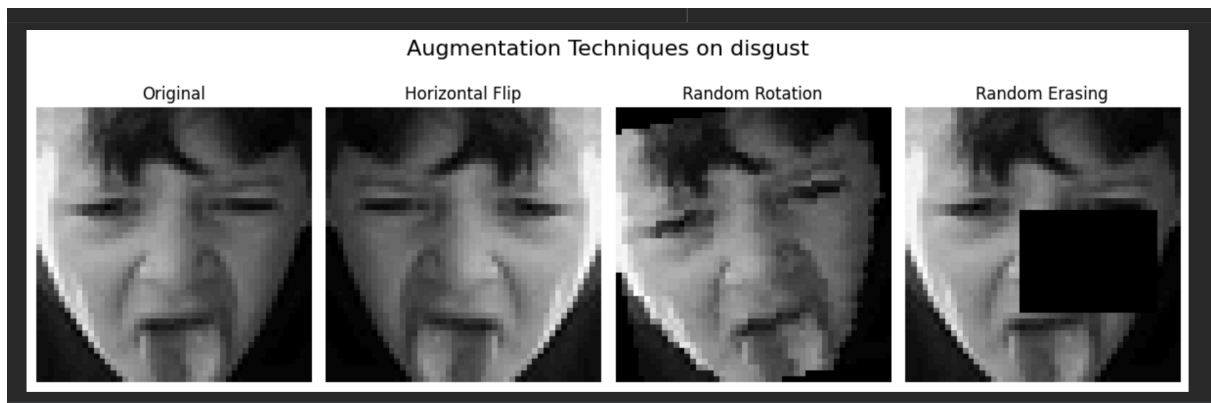




assignment 3

Status	Finale
Description	train a model and understand its predictions
Completion	@December 28, 2025
Design Tools	

using FER2013 facial expression dataset because.. i think it's interesting to first of all classify emotions into more scientific objectifiable truths when humans can misinterpret emotions all the time. im also really interested in micro-expressions so it could be a start into looking into catching these tiny realities of emotions that humans can barely pick up. does it become like a lie detector test? so i picked it because i wanted to see the way a machine could pick up emotions. and it has loads of robotic applications.



conclusion

- epochs had strongest effect of accuracy - 54% at 5 epochs to 91.88% at 45 epochs
- learning rate when too low, the model learnt so slow it became static
- batch size. the smaller the faster it learnt at the beginning but converge worse whereas larger batches learn slower but with better accuracy at the end
- data augmentation on training data distorted my dataset too much probably bc of pixels, beacuse of the low-resolution images and destroyed important features for facial expressions

FER2013 is a challenging dataset because the images are small, greyscale, and facial expressions are very subtle. I started with a simple CNN, batch size 128, learning rate 0.001, and horizontal flip augmentation to prevent bias with tilting faces.

First, I tested **epochs**. More epochs improved accuracy: 5 → 15 epochs increased from 54% to 69%, 23 epochs gave 76.7%, 35 epochs 82.6%, and 45 epochs reached 88.97%. Validation accuracy continued to improve, so the model was learning subtle features without significant overfitting.

Next, I tested **learning rate**. Lowering it to 0.0001 made training slower and accuracy dropped to 54%, showing that too small a learning rate prevents convergence.

Then, I tested **batch size**. Smaller batches (96 or 64) made training faster initially but reduced final accuracy. Larger batch size (128) gave slower but more stable learning and the best results after enough epochs.

Finally, I tested **augmentation**. Horizontal flip helped, but random rotation or random erasing was too destructive for FER2013, lowering accuracy. Minimal augmentation worked best.

- for speed: smaller batch, fewer epochs → faster early learning, lower final accuracy
- for best accuracy: larger batch, more epochs → slower start, better final convergence
- next i would want to decrease learning rate over time so it can keep learning and fine tuning..

TESTING

EPOCH TRAINING

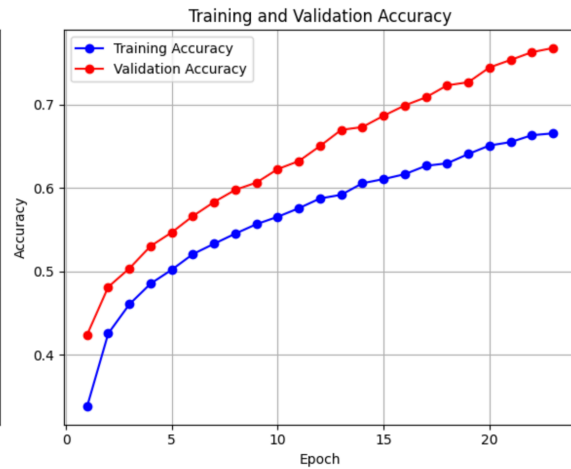
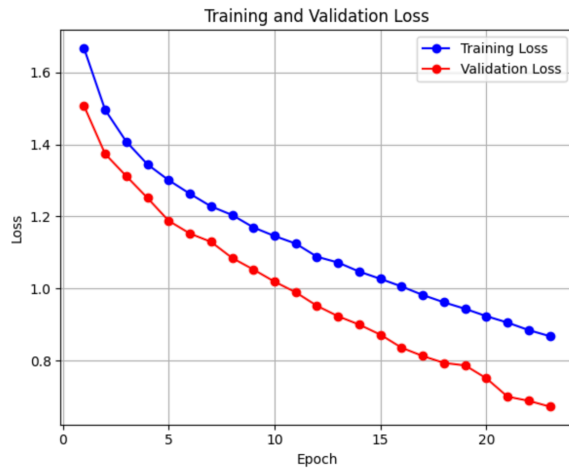
#test 1+2

- i have started with **54%** accuracy with 5 epochs
- gone up to **69.31%** with 15 epochs

→ shows im not giving it enough time to learn anything

#test 2.5

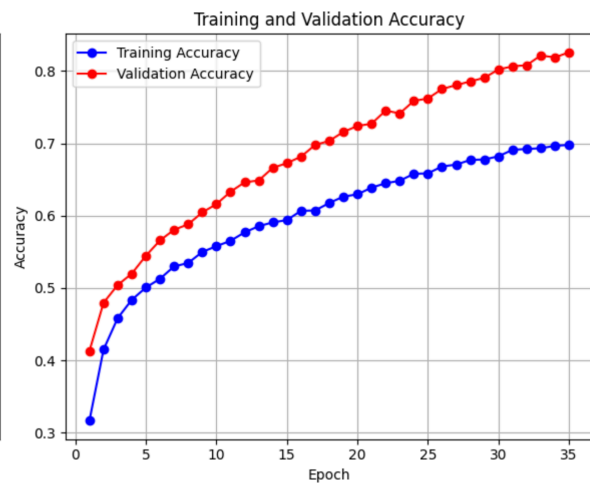
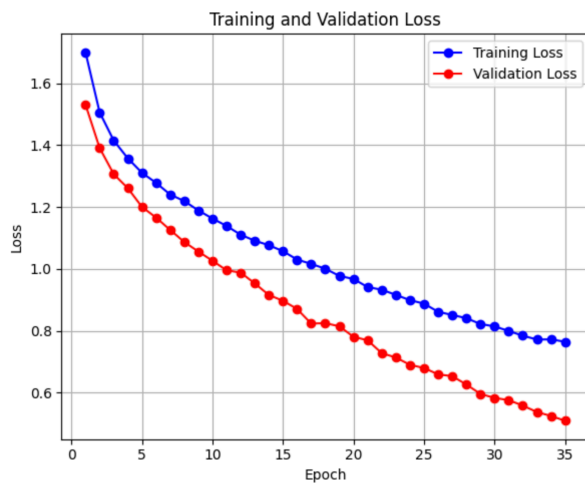
- 23 EPOCHS → **76.76%**



so the model is now learning useful facial features and not guessing.

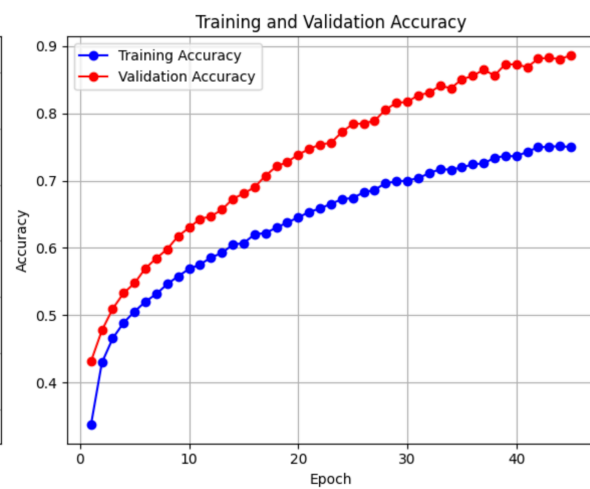
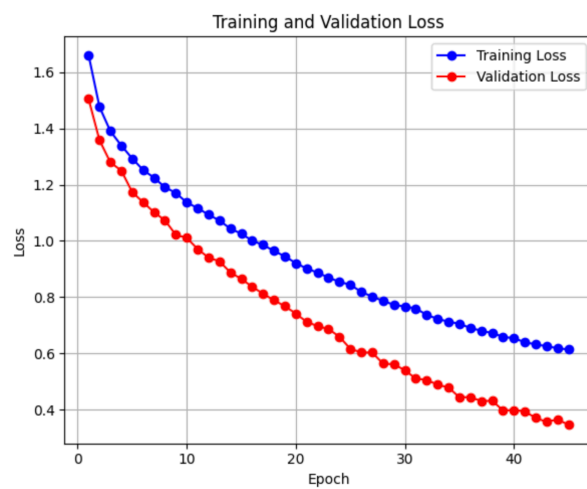
#test 2.6

35 epochs → **82.59%** (128 batch, 0.001 lr)



#test2.7

45 epochs → **88.67%** (128 batch, 0.001lr) **BEST RESULT SO FAR**



```

Epoch 30/45: Train Loss=0.7667, Train Acc=0.7001, Val Loss=0.5397, Val Acc=0.8171
Epoch 31/45: Train Loss=0.7597, Train Acc=0.7038, Val Loss=0.5104, Val Acc=0.8267
Epoch 32/45: Train Loss=0.7370, Train Acc=0.7110, Val Loss=0.5059, Val Acc=0.8306
Epoch 33/45: Train Loss=0.7227, Train Acc=0.7171, Val Loss=0.4885, Val Acc=0.8409
Epoch 34/45: Train Loss=0.7125, Train Acc=0.7164, Val Loss=0.4789, Val Acc=0.8375
Epoch 35/45: Train Loss=0.7054, Train Acc=0.7196, Val Loss=0.4435, Val Acc=0.8504
Epoch 36/45: Train Loss=0.6904, Train Acc=0.7243, Val Loss=0.4449, Val Acc=0.8558
Epoch 37/45: Train Loss=0.6808, Train Acc=0.7257, Val Loss=0.4302, Val Acc=0.8646
Epoch 38/45: Train Loss=0.6720, Train Acc=0.7340, Val Loss=0.4316, Val Acc=0.8564
Epoch 39/45: Train Loss=0.6593, Train Acc=0.7368, Val Loss=0.3971, Val Acc=0.8733
Epoch 40/45: Train Loss=0.6534, Train Acc=0.7362, Val Loss=0.3985, Val Acc=0.8726
Epoch 41/45: Train Loss=0.6399, Train Acc=0.7430, Val Loss=0.3935, Val Acc=0.8687
Epoch 42/45: Train Loss=0.6317, Train Acc=0.7496, Val Loss=0.3701, Val Acc=0.8812
Epoch 43/45: Train Loss=0.6258, Train Acc=0.7504, Val Loss=0.3563, Val Acc=0.8827
Epoch 44/45: Train Loss=0.6181, Train Acc=0.7514, Val Loss=0.3655, Val Acc=0.8804
Epoch 45/45: Train Loss=0.6140, Train Acc=0.7498, Val Loss=0.3456, Val Acc=0.8867

```

16 min

→ shows that the model needs time to learn subtle facial features, but it feels like it is starting to level out at end and could start overfitting soon - **more epochs = better learning**

LEARNING RATE TEST

#test 3

```

Epoch 1/23: Train Loss=1.7614, Train Acc=0.2809, Val Loss=1.6627, Val Acc=0.3494
Epoch 2/23: Train Loss=1.6590, Train Acc=0.3453, Val Loss=1.5985, Val Acc=0.3740
Epoch 3/23: Train Loss=1.6136, Train Acc=0.3712, Val Loss=1.5662, Val Acc=0.3902
Epoch 4/23: Train Loss=1.5728, Train Acc=0.3950, Val Loss=1.5124, Val Acc=0.4260
Epoch 5/23: Train Loss=1.5391, Train Acc=0.4080, Val Loss=1.4828, Val Acc=0.4336
Epoch 6/23: Train Loss=1.5119, Train Acc=0.4229, Val Loss=1.4568, Val Acc=0.4413
Epoch 7/23: Train Loss=1.4909, Train Acc=0.4273, Val Loss=1.4355, Val Acc=0.4533
Epoch 8/23: Train Loss=1.4646, Train Acc=0.4397, Val Loss=1.4063, Val Acc=0.4726
Epoch 9/23: Train Loss=1.4504, Train Acc=0.4481, Val Loss=1.3881, Val Acc=0.4740
Epoch 10/23: Train Loss=1.4308, Train Acc=0.4565, Val Loss=1.3740, Val Acc=0.4777
Epoch 11/23: Train Loss=1.4151, Train Acc=0.4579, Val Loss=1.3564, Val Acc=0.4883
Epoch 12/23: Train Loss=1.4013, Train Acc=0.4629, Val Loss=1.3487, Val Acc=0.4950
Epoch 13/23: Train Loss=1.3885, Train Acc=0.4703, Val Loss=1.3378, Val Acc=0.4993
Epoch 14/23: Train Loss=1.3801, Train Acc=0.4759, Val Loss=1.3134, Val Acc=0.5060
Epoch 15/23: Train Loss=1.3650, Train Acc=0.4828, Val Loss=1.3097, Val Acc=0.5067

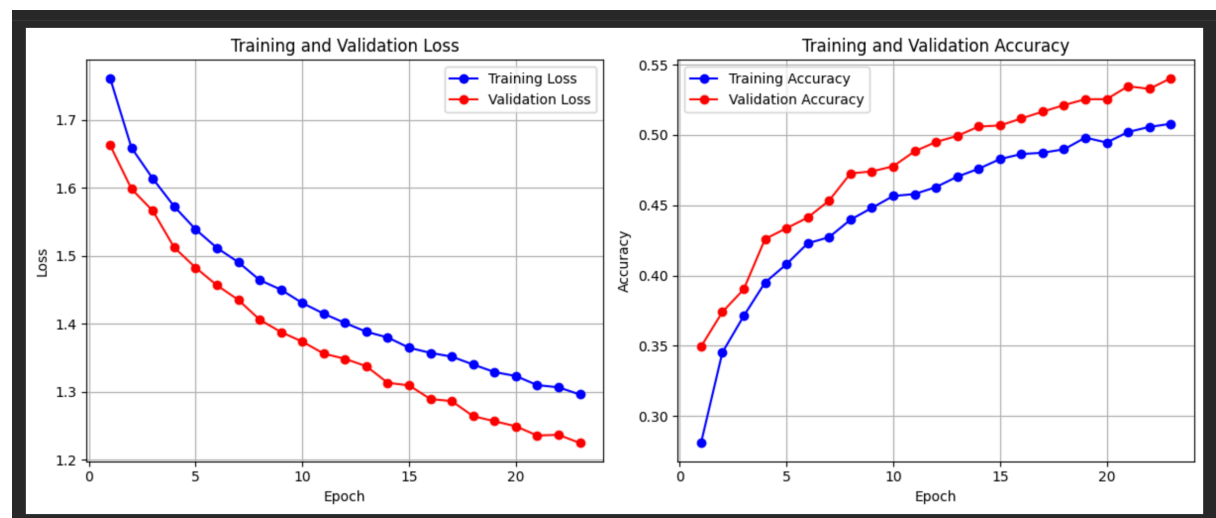
```

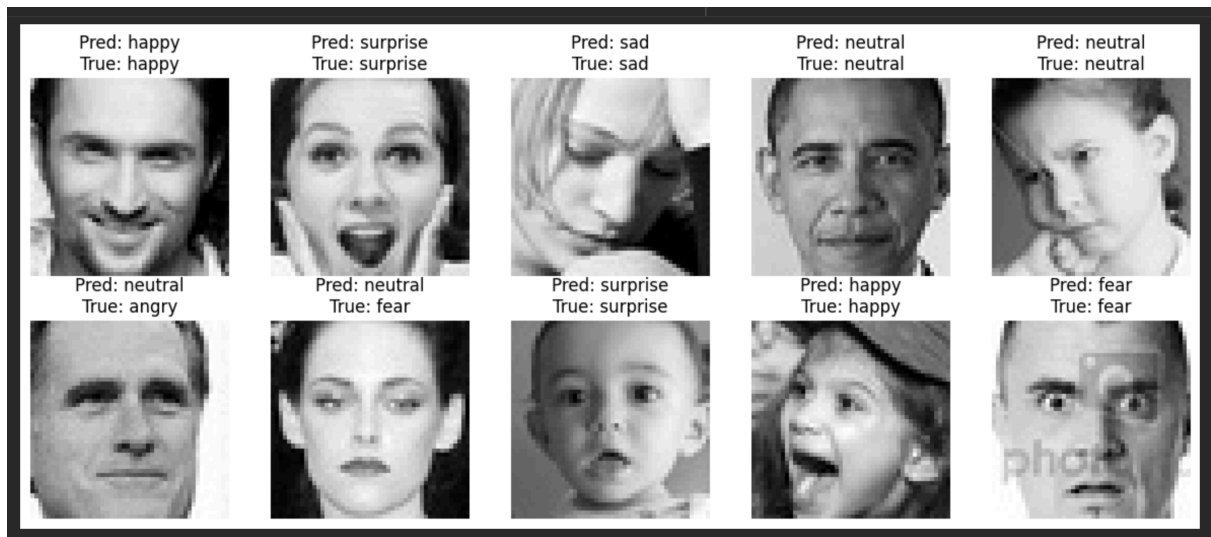
learning rate = 0.0001 (- decreased from 0.001)

23 epochs

→ feels slower, less accuracy = **54.04%**

0.0001 is too low for this dataset (lr too small basically static) - **learning too carefully = stuck**





→ though here in the samples you can see **how wrong the 'true' testing dataset is** - that is not fear or anger so i think a huge problem with my tests is how wrong the testing data is.

TESTING BATCH SIZE

#test 4

learning rate = 0.001 (+)

23 epochs (=)

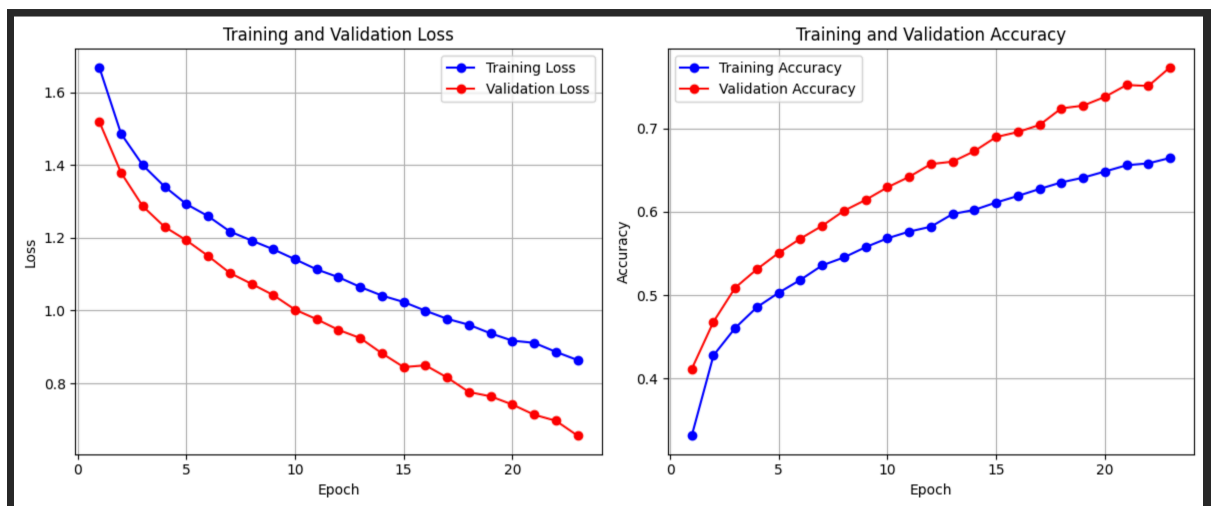
128 batch (=)

horizontal flip augmentation (=)

→ **77.34%**

→ 128 is stable but slower

```
Epoch 1/23: Train Loss=1.6683, Train Acc=0.3313, Val Loss=1.5189, Val Acc=0.4114
Epoch 2/23: Train Loss=1.4858, Train Acc=0.4277, Val Loss=1.3787, Val Acc=0.4680
Epoch 3/23: Train Loss=1.3993, Train Acc=0.4606, Val Loss=1.2863, Val Acc=0.5090
Epoch 4/23: Train Loss=1.3407, Train Acc=0.4855, Val Loss=1.2301, Val Acc=0.5311
Epoch 5/23: Train Loss=1.2925, Train Acc=0.5029, Val Loss=1.1931, Val Acc=0.5507
Epoch 6/23: Train Loss=1.2594, Train Acc=0.5183, Val Loss=1.1501, Val Acc=0.5678
Epoch 7/23: Train Loss=1.2173, Train Acc=0.5357, Val Loss=1.1033, Val Acc=0.5832
Epoch 8/23: Train Loss=1.1924, Train Acc=0.5452, Val Loss=1.0733, Val Acc=0.6012
Epoch 9/23: Train Loss=1.1682, Train Acc=0.5576, Val Loss=1.0424, Val Acc=0.6143
Epoch 10/23: Train Loss=1.1408, Train Acc=0.5682, Val Loss=1.0024, Val Acc=0.6294
Epoch 11/23: Train Loss=1.1129, Train Acc=0.5763, Val Loss=0.9760, Val Acc=0.6420
Epoch 12/23: Train Loss=1.0915, Train Acc=0.5821, Val Loss=0.9467, Val Acc=0.6573
Epoch 13/23: Train Loss=1.0647, Train Acc=0.5973, Val Loss=0.9243, Val Acc=0.6602
Epoch 14/23: Train Loss=1.0410, Train Acc=0.6023, Val Loss=0.8823, Val Acc=0.6728
Epoch 15/23: Train Loss=1.0233, Train Acc=0.6111, Val Loss=0.8445, Val Acc=0.6897
```



#test 5 // random cos 25 epochs instead of 23 so bit unhelpful

batch = 96 rather than 128 (-)

lr = 0.001 (=)

25 epochs (+)

78.18%

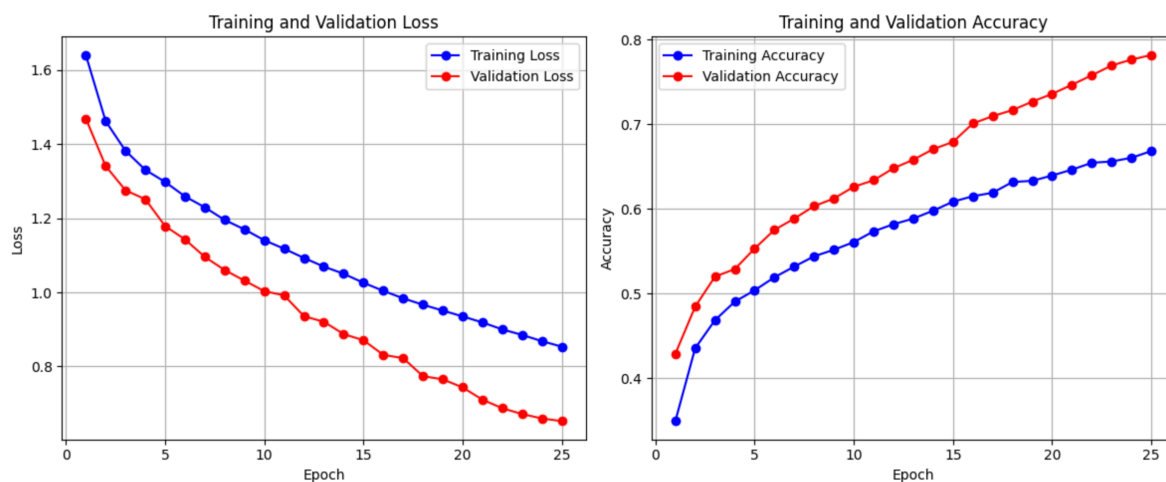
just testing smaller batch - still higher but does have 2 more epochs

→ faster improvement from the beginning and slightly better accuracy



i would also say that is 'sad' not 'fear'

Epoch 1/25: Train Loss=1.6409, Train Acc=0.3491, Val Loss=1.4692, Val Acc=0.4286
Epoch 2/25: Train Loss=1.4621, Train Acc=0.4352, Val Loss=1.3409, Val Acc=0.4844
Epoch 3/25: Train Loss=1.3822, Train Acc=0.4684, Val Loss=1.2756, Val Acc=0.5199
Epoch 4/25: Train Loss=1.3306, Train Acc=0.4903, Val Loss=1.2513, Val Acc=0.5288
Epoch 5/25: Train Loss=1.2980, Train Acc=0.5037, Val Loss=1.1788, Val Acc=0.5532
Epoch 6/25: Train Loss=1.2590, Train Acc=0.5191, Val Loss=1.1420, Val Acc=0.5752
Epoch 7/25: Train Loss=1.2294, Train Acc=0.5318, Val Loss=1.0956, Val Acc=0.5885
Epoch 8/25: Train Loss=1.1953, Train Acc=0.5439, Val Loss=1.0602, Val Acc=0.6032
Epoch 9/25: Train Loss=1.1699, Train Acc=0.5516, Val Loss=1.0316, Val Acc=0.6123
Epoch 10/25: Train Loss=1.1406, Train Acc=0.5607, Val Loss=1.0031, Val Acc=0.6261
Epoch 11/25: Train Loss=1.1176, Train Acc=0.5734, Val Loss=0.9918, Val Acc=0.6338
Epoch 12/25: Train Loss=1.0927, Train Acc=0.5818, Val Loss=0.9354, Val Acc=0.6480
Epoch 13/25: Train Loss=1.0695, Train Acc=0.5885, Val Loss=0.9206, Val Acc=0.6580
Epoch 14/25: Train Loss=1.0502, Train Acc=0.5978, Val Loss=0.8872, Val Acc=0.6706
Epoch 15/25: Train Loss=1.0258, Train Acc=0.6085, Val Loss=0.8706, Val Acc=0.6791



#test 6

next test: 96 batch size

same everything else

0.001 lr

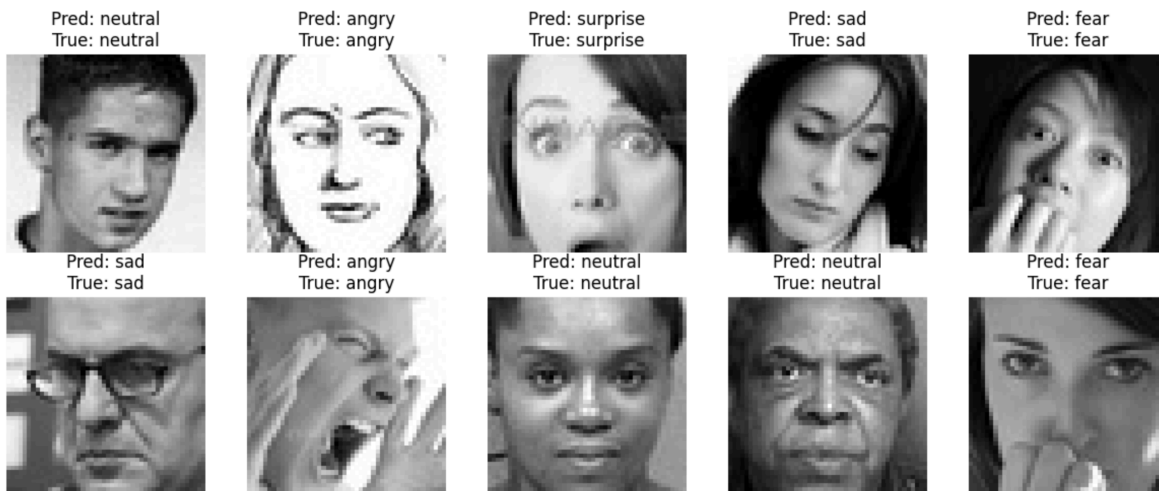
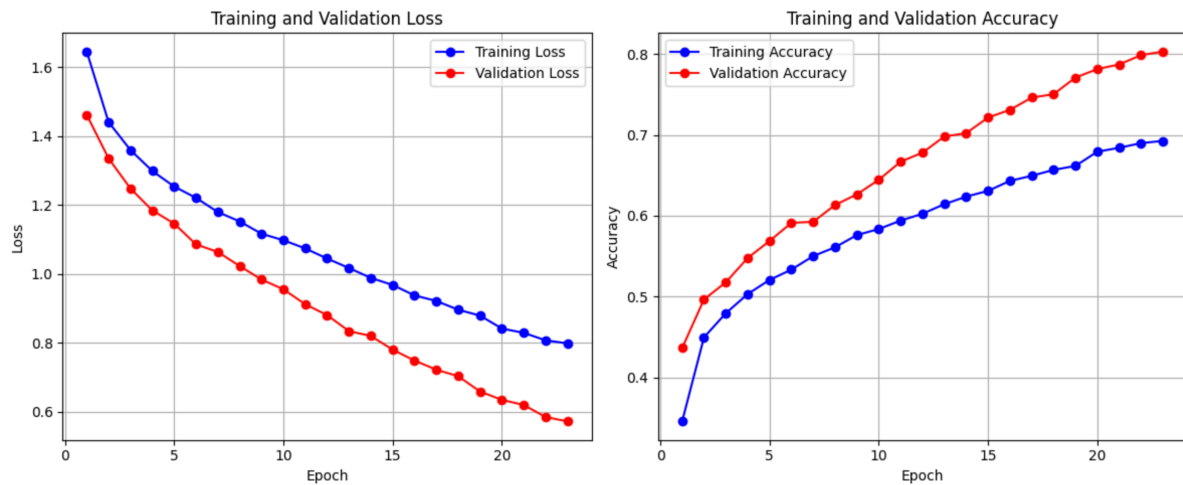
23 epochs

1 augmentation

→ **80.30% BEST RESULT - smaller batch**

SO: smaller batch size — but is same as test 6??

```
Epoch 1/23: Train Loss=1.6455, Train Acc=0.3456, Val Loss=1.4620, Val Acc=0.4363
Epoch 2/23: Train Loss=1.4409, Train Acc=0.4495, Val Loss=1.3355, Val Acc=0.4964
Epoch 3/23: Train Loss=1.3599, Train Acc=0.4794, Val Loss=1.2483, Val Acc=0.5177
Epoch 4/23: Train Loss=1.2991, Train Acc=0.5031, Val Loss=1.1849, Val Acc=0.5477
Epoch 5/23: Train Loss=1.2540, Train Acc=0.5206, Val Loss=1.1464, Val Acc=0.5687
Epoch 6/23: Train Loss=1.2210, Train Acc=0.5333, Val Loss=1.0862, Val Acc=0.5911
Epoch 7/23: Train Loss=1.1800, Train Acc=0.5501, Val Loss=1.0641, Val Acc=0.5925
Epoch 8/23: Train Loss=1.1527, Train Acc=0.5609, Val Loss=1.0233, Val Acc=0.6133
Epoch 9/23: Train Loss=1.1169, Train Acc=0.5761, Val Loss=0.9842, Val Acc=0.6264
Epoch 10/23: Train Loss=1.0982, Train Acc=0.5835, Val Loss=0.9559, Val Acc=0.6444
Epoch 11/23: Train Loss=1.0744, Train Acc=0.5940, Val Loss=0.9126, Val Acc=0.6668
Epoch 12/23: Train Loss=1.0451, Train Acc=0.6024, Val Loss=0.8810, Val Acc=0.6780
Epoch 13/23: Train Loss=1.0176, Train Acc=0.6142, Val Loss=0.8347, Val Acc=0.6979
Epoch 14/23: Train Loss=0.9890, Train Acc=0.6236, Val Loss=0.8205, Val Acc=0.7021
Epoch 15/23: Train Loss=0.9684, Train Acc=0.6307, Val Loss=0.7808, Val Acc=0.7214
Epoch 16/23: Train Loss=0.9382, Train Acc=0.6430, Val Loss=0.7491, Val Acc=0.7309
Epoch 17/23: Train Loss=0.9220, Train Acc=0.6493, Val Loss=0.7227, Val Acc=0.7461
Epoch 18/23: Train Loss=0.8966, Train Acc=0.6567, Val Loss=0.7037, Val Acc=0.7503
Epoch 19/23: Train Loss=0.8796, Train Acc=0.6615, Val Loss=0.6594, Val Acc=0.7708
Epoch 20/23: Train Loss=0.8420, Train Acc=0.6792, Val Loss=0.6352, Val Acc=0.7813
Epoch 21/23: Train Loss=0.8295, Train Acc=0.6840, Val Loss=0.6203, Val Acc=0.7870
Epoch 22/23: Train Loss=0.8079, Train Acc=0.6897, Val Loss=0.5854, Val Acc=0.7985
Epoch 23/23: Train Loss=0.7990, Train Acc=0.6925, Val Loss=0.5727, Val Acc=0.8030
Training complete.
```



so far → shows stable convergence with no overfitting

TESTING AUGMENTATION

#test 7

ADDING AUGMENTATION TO DATA

so now add more than

```
transforms.RandomHorizontalFlip(p=0.5), #so it doesn't have dataset biases like tilting heads
```

adding **random rotation** and **random erasing**:

```
# Define simple augmentations for training
train_transform = transforms.Compose([
    transforms.Grayscale(num_output_channels=1), #transforms applied per dataset instantiation
    transforms.Resize((48,48)), #resize to img sizes
    transforms.RandomHorizontalFlip(p=0.5), #so it doesn't have dataset biases like tilting heads
    transforms.RandomRotation(degrees=15),
    transforms.ToTensor(),
    transforms.RandomErasing(p=0.5, scale=(0.02, 0.2)),
    transforms.Normalize(mean=[0.5], std=[0.5]) #normalise pixel values ??????
])
```

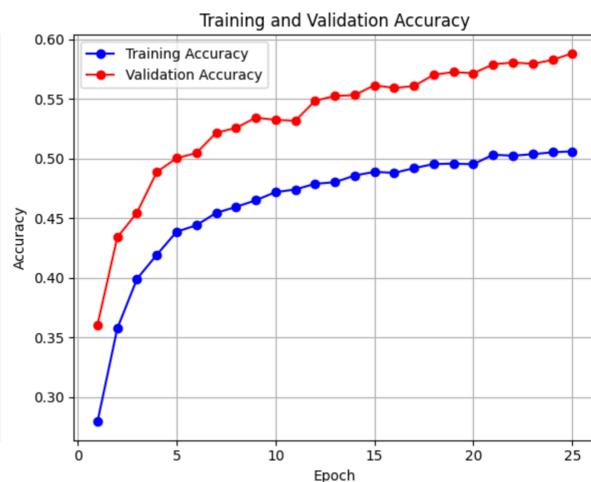
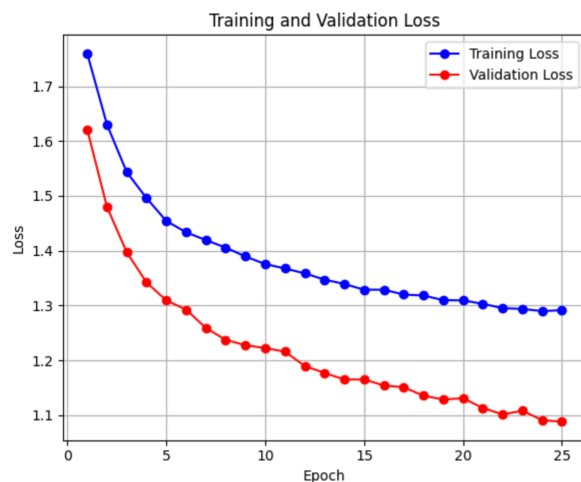
has to be after .tensortensor bc it is tensor-only transformation, not PIL image transformation

maybe too high 0.5 and 15 degrees?

→ **58.82% TERRIBLE**

→ **60% 35 epochs**

```
Epoch 1/25: Train Loss=1.7599, Train Acc=0.2794, Val Loss=1.6214, Val Acc=0.3606
Epoch 2/25: Train Loss=1.6301, Train Acc=0.3575, Val Loss=1.4801, Val Acc=0.4339
Epoch 3/25: Train Loss=1.5433, Train Acc=0.3992, Val Loss=1.3972, Val Acc=0.4548
Epoch 4/25: Train Loss=1.4959, Train Acc=0.4194, Val Loss=1.3422, Val Acc=0.4888
Epoch 5/25: Train Loss=1.4543, Train Acc=0.4387, Val Loss=1.3096, Val Acc=0.5003
Epoch 6/25: Train Loss=1.4336, Train Acc=0.4441, Val Loss=1.2927, Val Acc=0.5047
Epoch 7/25: Train Loss=1.4194, Train Acc=0.4545, Val Loss=1.2594, Val Acc=0.5217
Epoch 8/25: Train Loss=1.4058, Train Acc=0.4595, Val Loss=1.2376, Val Acc=0.5257
Epoch 9/25: Train Loss=1.3895, Train Acc=0.4650, Val Loss=1.2279, Val Acc=0.5343
Epoch 10/25: Train Loss=1.3757, Train Acc=0.4719, Val Loss=1.2226, Val Acc=0.5324
Epoch 11/25: Train Loss=1.3677, Train Acc=0.4741, Val Loss=1.2160, Val Acc=0.5317
Epoch 12/25: Train Loss=1.3587, Train Acc=0.4789, Val Loss=1.1901, Val Acc=0.5485
Epoch 13/25: Train Loss=1.3473, Train Acc=0.4802, Val Loss=1.1773, Val Acc=0.5525
Epoch 14/25: Train Loss=1.3394, Train Acc=0.4858, Val Loss=1.1654, Val Acc=0.5532
Epoch 15/25: Train Loss=1.3290, Train Acc=0.4889, Val Loss=1.1653, Val Acc=0.5615
Epoch 16/25: Train Loss=1.3290, Train Acc=0.4880, Val Loss=1.1544, Val Acc=0.5592
Epoch 17/25: Train Loss=1.3201, Train Acc=0.4920, Val Loss=1.1512, Val Acc=0.5608
Epoch 18/25: Train Loss=1.3184, Train Acc=0.4954, Val Loss=1.1358, Val Acc=0.5703
Epoch 19/25: Train Loss=1.3100, Train Acc=0.4958, Val Loss=1.1288, Val Acc=0.5725
Epoch 20/25: Train Loss=1.3095, Train Acc=0.4952, Val Loss=1.1312, Val Acc=0.5715
Epoch 21/25: Train Loss=1.3032, Train Acc=0.5033, Val Loss=1.1134, Val Acc=0.5790
Epoch 22/25: Train Loss=1.2954, Train Acc=0.5024, Val Loss=1.1017, Val Acc=0.5805
Epoch 23/25: Train Loss=1.2939, Train Acc=0.5037, Val Loss=1.1084, Val Acc=0.5794
Epoch 24/25: Train Loss=1.2900, Train Acc=0.5053, Val Loss=1.0913, Val Acc=0.5827
Epoch 25/25: Train Loss=1.2916, Train Acc=0.5061, Val Loss=1.0884, Val Acc=0.5882
Training complete.
```



→ **too destructive for my dataset**

becomes distorted images and corrupt dataset rather than helpful for training

#test 8

adding more epochs - 35 - when adding data augmentation

and lowering probabilities and rotation - augmenting less

to help generalisation

→ **63.05%**

- still decreasing much slower with more augmentation

so increased because i augmented less but still too much augmentation on this sample of data

#test 9

35 epochs

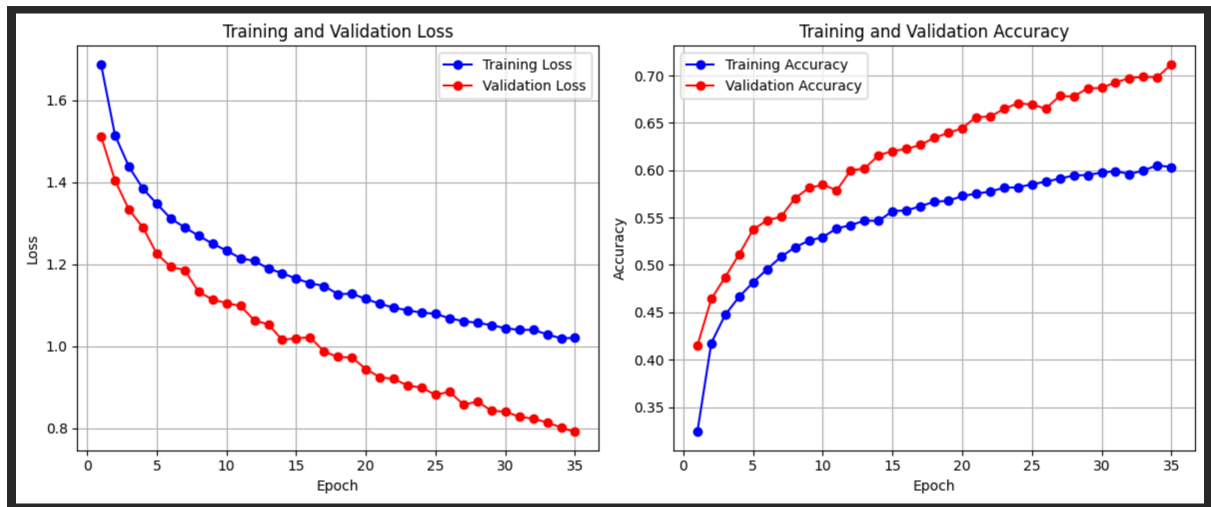
0.001 lr

96 batch

horizontal flip,

random rotation 5 degrees. removed erasing

→ 71.21%



— end of augmentation tests

#test 10 — TESTING BATCH SIZE TOO

35 epochs

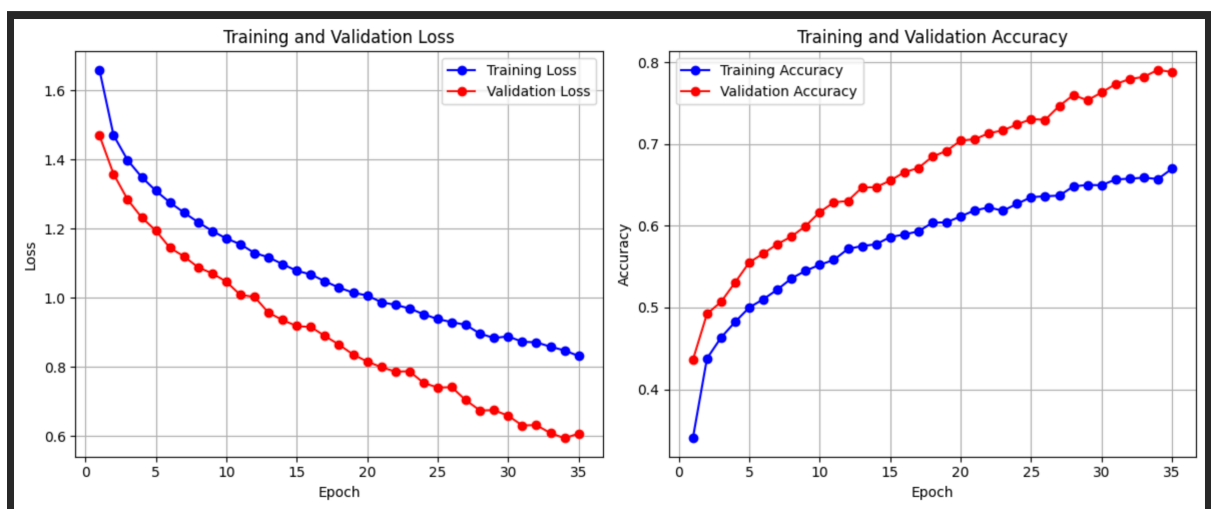
64 batch - smaller batch size

just horizontal flip augmentation. removed rotation bc pixel distortion

0.001 lr

→ 78.79%

→ ??




```

Epoch 1/35: Train Loss=1.6590, Train Acc=0.3402, Val Loss=1.4695, Val Acc=0.4359
Epoch 2/35: Train Loss=1.4699, Train Acc=0.4376, Val Loss=1.3579, Val Acc=0.4921
Epoch 3/35: Train Loss=1.3977, Train Acc=0.4632, Val Loss=1.2838, Val Acc=0.5069
Epoch 4/35: Train Loss=1.3489, Train Acc=0.4824, Val Loss=1.2322, Val Acc=0.5307
Epoch 5/35: Train Loss=1.3095, Train Acc=0.5000, Val Loss=1.1944, Val Acc=0.5551
Epoch 6/35: Train Loss=1.2756, Train Acc=0.5101, Val Loss=1.1446, Val Acc=0.5661
Epoch 7/35: Train Loss=1.2465, Train Acc=0.5220, Val Loss=1.1198, Val Acc=0.5776
Epoch 8/35: Train Loss=1.2186, Train Acc=0.5356, Val Loss=1.0882, Val Acc=0.5869
Epoch 9/35: Train Loss=1.1932, Train Acc=0.5450, Val Loss=1.0708, Val Acc=0.5997
Epoch 10/35: Train Loss=1.1718, Train Acc=0.5522, Val Loss=1.0469, Val Acc=0.6166
Epoch 11/35: Train Loss=1.1549, Train Acc=0.5582, Val Loss=1.0084, Val Acc=0.6286
Epoch 12/35: Train Loss=1.1297, Train Acc=0.5716, Val Loss=1.0028, Val Acc=0.6303
Epoch 13/35: Train Loss=1.1172, Train Acc=0.5752, Val Loss=0.9569, Val Acc=0.6468
Epoch 14/35: Train Loss=1.0972, Train Acc=0.5773, Val Loss=0.9359, Val Acc=0.6472
Epoch 15/35: Train Loss=1.0782, Train Acc=0.5862, Val Loss=0.9185, Val Acc=0.6552
Epoch 16/35: Train Loss=1.0678, Train Acc=0.5893, Val Loss=0.9158, Val Acc=0.6654
Epoch 17/35: Train Loss=1.0477, Train Acc=0.5931, Val Loss=0.8898, Val Acc=0.6707
Epoch 18/35: Train Loss=1.0296, Train Acc=0.6037, Val Loss=0.8647, Val Acc=0.6845
Epoch 19/35: Train Loss=1.0150, Train Acc=0.6042, Val Loss=0.8364, Val Acc=0.6916
Epoch 20/35: Train Loss=1.0069, Train Acc=0.6114, Val Loss=0.8161, Val Acc=0.7042
Epoch 21/35: Train Loss=0.9876, Train Acc=0.6188, Val Loss=0.7997, Val Acc=0.7058
Epoch 22/35: Train Loss=0.9798, Train Acc=0.6224, Val Loss=0.7869, Val Acc=0.7131
Epoch 23/35: Train Loss=0.9695, Train Acc=0.6183, Val Loss=0.7879, Val Acc=0.7167
Epoch 24/35: Train Loss=0.9527, Train Acc=0.6268, Val Loss=0.7541, Val Acc=0.7238
Epoch 25/35: Train Loss=0.9387, Train Acc=0.6349, Val Loss=0.7410, Val Acc=0.7305
Epoch 26/35: Train Loss=0.9299, Train Acc=0.6361, Val Loss=0.7418, Val Acc=0.7296
Epoch 27/35: Train Loss=0.9219, Train Acc=0.6368, Val Loss=0.7042, Val Acc=0.7465
Epoch 28/35: Train Loss=0.8966, Train Acc=0.6481, Val Loss=0.6747, Val Acc=0.7599
Epoch 29/35: Train Loss=0.8841, Train Acc=0.6499, Val Loss=0.6755, Val Acc=0.7534
Epoch 30/35: Train Loss=0.8886, Train Acc=0.6496, Val Loss=0.6594, Val Acc=0.7630
Epoch 31/35: Train Loss=0.8731, Train Acc=0.6566, Val Loss=0.6314, Val Acc=0.7733
Epoch 32/35: Train Loss=0.8715, Train Acc=0.6576, Val Loss=0.6323, Val Acc=0.7794
Epoch 33/35: Train Loss=0.8589, Train Acc=0.6588, Val Loss=0.6093, Val Acc=0.7821
Epoch 34/35: Train Loss=0.8481, Train Acc=0.6571, Val Loss=0.5945, Val Acc=0.7907
Epoch 35/35: Train Loss=0.8321, Train Acc=0.6697, Val Loss=0.6077, Val Acc=0.7879
Training complete.

```

→ augmentation only helps when the data can be distorted but in terms of **fer2013** it **can't** due to the **small pixelated pictures and the subtle facial expressions it needs to learn**.

#LAST TEST 11

128 batch size

0.001 lr

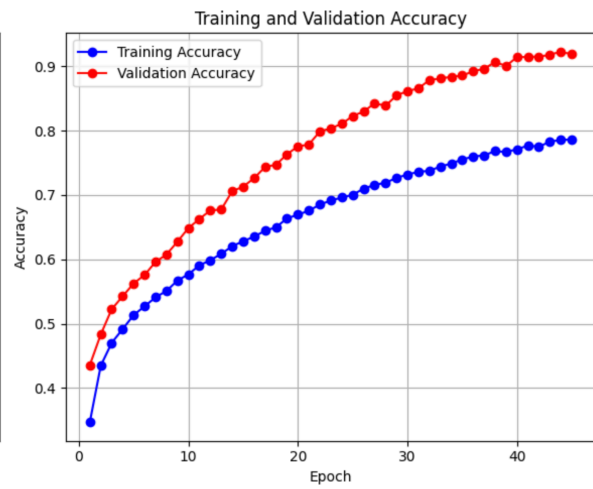
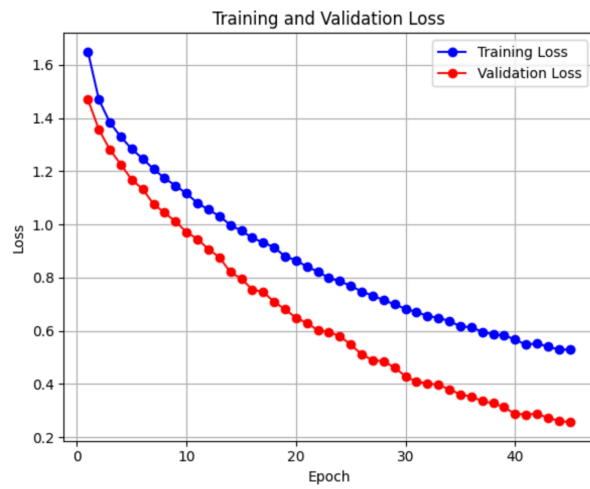
45 epochs

→ 91.88%

```

Epoch 15/45: Train Loss=0.9762, Train Acc=0.6281, Val Loss=0.7962, Val Acc=0.7131
Epoch 16/45: Train Loss=0.9501, Train Acc=0.6354, Val Loss=0.7553, Val Acc=0.7262
Epoch 17/45: Train Loss=0.9316, Train Acc=0.6444, Val Loss=0.7462, Val Acc=0.7437
Epoch 18/45: Train Loss=0.9147, Train Acc=0.6499, Val Loss=0.7089, Val Acc=0.7472
Epoch 19/45: Train Loss=0.8791, Train Acc=0.6637, Val Loss=0.6809, Val Acc=0.7633
Epoch 20/45: Train Loss=0.8657, Train Acc=0.6694, Val Loss=0.6490, Val Acc=0.7757
Epoch 21/45: Train Loss=0.8406, Train Acc=0.6757, Val Loss=0.6296, Val Acc=0.7786
Epoch 22/45: Train Loss=0.8233, Train Acc=0.6851, Val Loss=0.6021, Val Acc=0.7988
Epoch 23/45: Train Loss=0.7979, Train Acc=0.6915, Val Loss=0.5957, Val Acc=0.8038
Epoch 24/45: Train Loss=0.7871, Train Acc=0.6961, Val Loss=0.5798, Val Acc=0.8107
Epoch 25/45: Train Loss=0.7685, Train Acc=0.7004, Val Loss=0.5480, Val Acc=0.8229
Epoch 26/45: Train Loss=0.7480, Train Acc=0.7100, Val Loss=0.5116, Val Acc=0.8299
Epoch 27/45: Train Loss=0.7319, Train Acc=0.7157, Val Loss=0.4898, Val Acc=0.8426
Epoch 28/45: Train Loss=0.7157, Train Acc=0.7192, Val Loss=0.4856, Val Acc=0.8389
Epoch 29/45: Train Loss=0.6993, Train Acc=0.7265, Val Loss=0.4621, Val Acc=0.8549
Epoch 30/45: Train Loss=0.6828, Train Acc=0.7320, Val Loss=0.4288, Val Acc=0.8618
Epoch 31/45: Train Loss=0.6702, Train Acc=0.7360, Val Loss=0.4092, Val Acc=0.8658
Epoch 32/45: Train Loss=0.6565, Train Acc=0.7377, Val Loss=0.4032, Val Acc=0.8788
Epoch 33/45: Train Loss=0.6486, Train Acc=0.7442, Val Loss=0.3968, Val Acc=0.8812
Epoch 34/45: Train Loss=0.6371, Train Acc=0.7490, Val Loss=0.3804, Val Acc=0.8840
Epoch 35/45: Train Loss=0.6168, Train Acc=0.7553, Val Loss=0.3609, Val Acc=0.8859
Epoch 36/45: Train Loss=0.6135, Train Acc=0.7598, Val Loss=0.3533, Val Acc=0.8924
Epoch 37/45: Train Loss=0.5968, Train Acc=0.7618, Val Loss=0.3359, Val Acc=0.8965
Epoch 38/45: Train Loss=0.5873, Train Acc=0.7682, Val Loss=0.3278, Val Acc=0.9068
Epoch 39/45: Train Loss=0.5831, Train Acc=0.7670, Val Loss=0.3131, Val Acc=0.9009
Epoch 40/45: Train Loss=0.5675, Train Acc=0.7711, Val Loss=0.2874, Val Acc=0.9143
Epoch 41/45: Train Loss=0.5476, Train Acc=0.7770, Val Loss=0.2860, Val Acc=0.9150
Epoch 42/45: Train Loss=0.5535, Train Acc=0.7752, Val Loss=0.2876, Val Acc=0.9151
Epoch 43/45: Train Loss=0.5400, Train Acc=0.7829, Val Loss=0.2733, Val Acc=0.9181
Epoch 44/45: Train Loss=0.5303, Train Acc=0.7857, Val Loss=0.2619, Val Acc=0.9231
Epoch 45/45: Train Loss=0.5287, Train Acc=0.7863, Val Loss=0.2564, Val Acc=0.9188
Training complete.

```



started levelling out and possibly start overfitting at the end - the validation accuracy.

the rate at which the model learns gets really slow. so would want to decrease learning rate over time so it can keep learning and fine tuning..